

Power BI Assignment 1 – Data Transformation and Data Modeling

Importing the data

1. Loaded List of Orders.csv using Get data > Text/CSV and clicked Transform Data to open Power Query.
2. Imported Order Details.csv and Sales target.csv from New Source in Power Query.

Data transformation

1. Kept only the first 500 rows of List of Orders (this also removed 60 blank rows).
2. Changed Order Date to Date type using Change Type > Using Locale, because the dates were in dd-mm-yyyy format.
3. Changed Amount and Target to Fixed Decimal Number.
4. Trimmed the text columns and used Capitalize Each Word on CustomerName.
5. Merged City and State into a new column called Location (City, State).
6. Added a custom column Profit Margin = Profit / Amount.
7. Added a conditional column Profit Status: Loss if profit is below 0, Break-Even if 0, and Profit if above 0.

Merging tables

Merged List of Orders and Order Details on Order ID (left outer join) into a new table called Orders Data.

Missing and duplicate data

The only missing data was 60 empty rows in List of Orders, which were removed when I kept the first 500 rows. I checked for duplicates using Remove Duplicates and found none.

Sorting and filtering

In Orders Data I sorted Order Date in descending order and filtered State to Tamil Nadu.

Grouping and aggregating

Using duplicate copies of the tables and Group By, I found the count of each Order ID, average profit by Category, total amount by Sub-Category, and total target by month.

Data modeling

1. Clicked Close & Apply and opened Model view.
2. Created a relationship between List of Orders and Order Details using Order ID.
3. Created a relationship between Order Details and Sales Target using Category.
4. Deleted the extra relationships Power BI created automatically, and checked in Manage relationships that both remaining relationships are active.